Contents

1 NeuroBOLT research directions **1**

1.1 1. Executive summary: fix evaluation, then test shared and lag-aware models 1

1.2 2. Fix before comparing scores: current numbers are not exact paper parity 2

1.3 3. A practical three-stage plan 3

1.4 4. Cutting-edge literature organized by research idea 4

1.5 5. First experiments on M3, then work that needs more data 5

1.6 6. Limitations and primary references 6

1.7 Conclusion 6

1 NeuroBOLT research directions

NeuroBOLT's most valuable next improvements are evaluation integrity and a small set of controlled, interpretable comparisons—not a larger model first. The repository's current cross-subject run differs from the paper in EEG filtering and AdamW β_2, and it drops held-out scan identities before metric aggregation; its temporal-split path also filters and normalizes across the full scan before splitting. Those facts do not prove the paper's results are invalid, but they do limit paper-parity and prospective-generalization claims. Keep the current M3 jobs running and label their output as current-code results with the deviations below, then run a corrected, scan-aware comparison. Once the baseline is trustworthy, the strongest code research directions are a shared multi-ROI decoder with functional-connectivity evaluation and a modest lag-aware EEG-to-BOLD model; newer EEG foundation models are worthwhile controls, but their published classification gains do not establish fMRI regression gains.

1.1 1. Executive summary: fix evaluation, then test shared and lag-aware models

The recommended order is to preserve the active run, repair evaluation and metadata handling for future runs, complete a paper-parity comparison, and only then spend GPU time on architecture changes. The highest-value science question is whether one model can predict multiple regions and preserve useful network structure across unseen people. The second is whether an inspectable lagged or region-conditioned temporal map improves on the current 16-second-window-to-one-frame prediction. Recent papers motivate both questions, but the local cohort has only 22 people and five held-out test participants; every claimed improvement therefore needs subject-level reporting and restrained interpretation.

1. Preserve scan, participant, frame, and split IDs; report scan-level and subject-level metrics; select checkpoints using validation only and evaluate test once.
Value: Very high | **Effort:** Low–medium | **Risk:** Low scientific risk; summary scores may change because aggregation becomes correct.

2. Save an explicit run manifest and add preflight checks for event/frame alignment, split grouping, finite inputs, and preprocessing settings.
Value: Very high | **Effort:** Low | **Risk:** Low; catches silent data and configuration errors.

3. Run a distinct paper-parity comparison with no extra EEG high-pass, AdamW β=(0.9, 0.99), and the same paper split/training schedule.
Value: Very high for reproduction | **Effort:** Low | **Risk:** Low; new outputs must be labeled separately from the active run.

4. Add transparent baselines, then compare seven independent ROI models with one shared seven-output model and scan-wise functional-connectivity metrics.
Value: High | **Effort:** Medium | **Risk:** Moderate; shared learning can cause negative transfer with few subjects.

5. Ablate lag/window choices and an interpretable temporal or HRF-inspired head while fixing all other settings.
Value: High mechanistic value | **Effort:** Medium | **Risk:** Moderate–high; HRF assumptions may be wrong for resting data or vary by region/state.

6. Compare a newer montage-aware pretrained encoder or adaptive spectral fusion only after baselines and evaluation are stable.
Value: Medium–high potential | **Effort:** Medium–high | **Risk:** High; classification benchmarks do not predict continuous BOLD regression performance.

Here, "value" means value for trustworthy claims or a meaningful research question, not an expected score increase. A small ridge model that establishes what the current Transformer adds is more useful than a larger model judged on a test set repeatedly inspected during development.

1.2 2. Fix before comparing scores: current numbers are not exact paper parity

Confirmed code facts. The current M3 cross-subject path uses prepare_full_dataloader, which applies a 0.5-Hz high-pass to EEG. NeuroBOLT's main reported setting does not add EEG high- or low-pass filtering; the paper describes 0.5-Hz high-pass as a sensitivity check that was not retained because results were close. The current path therefore represents the sensitivity variant, not the paper's primary EEG preprocessing. The paper's Table 5 reports AdamW β=(0.9, 0.99), whereas main.py leaves –opt_betas unset and optim_factory.py consequently uses PyTorch AdamW's β_2=0.999 default. The current M3 script does not override those betas. The paper and code details are in NeuroBOLT, the repository, and PyTorch's AdamW documentation.

The inter-scan reporting path loses the unit the paper reports. The full-test loader concatenates held-out scans into one dataset without retaining scan IDs. The training code then approximates six test batches by setting batch size from the total sample count divided by six; those chunks need not match the six held-out scan boundaries. The evaluator averages chunk correlations when it retains chunked predictions, rather than returning six scan-specific correlations and their standard deviation. The current collector also treats the inter-scan run as a single expected result. Consequently, the run cannot report the paper's per-held-out-scan mean and SD faithfully, even if its displayed mean correlation is numerically similar. Carry scan and subject IDs with each frame, compute each scan's R, MSE, and MAE independently, then report macro-averages and SD across scans and subjects. The paper's inter-subject table and the implementation locations are described in the paper and the checked-in dataset_maker/get_datasets.py, main.py, engine.py, utils.py, and scripts/collect_reproduction_results.py.

The test set is visible during training and affects checkpoint gating. main.py evaluates validation and test each epoch and saves a validation-MSE-best checkpoint only when test_corr > 0. The separate collector's choice of the minimum-validation-MSE epoch is validation-based, so the test score did not directly choose that row or update gradients. Still, test visibility at every epoch and a test-dependent save condition create an avoidable route for adaptive decisions and can prevent a checkpoint from being written. For confirmatory runs, save by validation criteria alone, freeze the checkpoint and configuration, then run the test set once. Treat this as an evaluation hygiene correction, not evidence that the current model trained on test labels.

Whole-scan target operations complicate temporal holdout claims. In the chronological intra-scan path, the BOLD target is filtered over the full scan with zero-phase filtfilt and normalized before the 80:10:10 temporal split. The normalization uses whole-scan statistics, while forward-backward filtering can use observations across split boundaries. In the cross-scan path, each held-out scan's target is also normalized from that scan's full time series before evaluation. These choices follow a per-scan preprocessing recipe but make the score transductive with respect to target scale; they are not a clean prospective test in which future BOLD is unavailable. SciPy documents the bidirectional operation in filtfilt. Keep a paper-faithful result for reproduction, and separately label a strict temporal-holdout variant that estimates target scaling only from training data and uses a causal or split-isolated filter.

Additional confirmed differences and checks. The single-scan epoch path applies MNE baseline correction over the entire 16-second EEG epoch; the paper's stated primary recipe does not describe this step. Its effect is unknown, so ablate the current setting against no epoch baseline correction rather than asserting it harms performance (MNE baseline documentation). Event-to-volume alignment is assumed in the preprocessing helper rather than fully checked; add assertions for event count, fMRI frame count, TR-like spacing, dropped initial frames, and selected indices. The split workbook is expected to group repeated scans by participant, but the loader does not enforce that condition.

Finally, the spectral encoder appears to omit the CLS token described for that block in the paper; this is a lower-priority exact-architecture discrepancy than the evaluation issues.

The current M3 jobs remain running; we deliberately did not alter them. Do not interrupt or reconfigure those jobs. When they finish, preserve their logs and outputs under a label such as "current code: 0.5-Hz high-pass, AdamW β_2=0.999, existing evaluation path." Follow them with a separately named corrected run; do not relabel the in-progress outputs as exact paper reproduction.

1.3 3. A practical three-stage plan

1.3.1 Stage 1 — integrity fixes for all future runs

Once the active array is done, make the result path scan-aware before interpreting architecture comparisons. Carry participant, scan, frame, ROI, and split IDs alongside the tensors; compute scan-level metrics before aggregating; and make the test evaluator a one-shot final step after validation selects the epoch. Remove test-dependent checkpoint gating in future runs. Add preflight checks for one split assignment per scan, no participant crossing split boundaries, event/frame correspondence, expected channel and target shapes, finite values, and an explicit manifest recording code revision, checkpoint hash, split workbook, ROI, preprocessing, optimizer betas, seed, and run label.

Expose the currently implicit choices as configuration fields: EEG high-pass, epoch baseline, target normalization, filter mode, and temporal split gap. The manifest should say whether a run is paper-faithful, strict prospective, or an exploratory ablation. Do not silently combine results across those protocols.

1.3.2 Stage 2 — low-risk baselines and ablations on the existing 29 scans

Use the existing 18/5/6 scan split and keep all scans from a participant together. First establish an intercept or training-mean baseline, then regularized ridge on lagged raw EEG and on log band-power or band-envelope features. Add a small EEGNet/TCN-style model if useful, and compare each against NeuroBOLT under the same preprocessing and scan-aware metrics. A previous-BOLD/AR(1) predictor can diagnose how much apparent predictability comes from smooth BOLD autocorrelation, but it is not an EEG-only deployment baseline. Add circular-shift or phase-randomized EEG nulls that preserve temporal structure while breaking EEG–BOLD alignment; do not independently shuffle overlapping windows.

Then test one factor at a time: paper-parity filtering and betas; baseline correction on/off; 8/16/24/32-second look-backs and prespecified target lags; seven separate ROI fits versus one shared seven-output model; and current equal multiscale fusion versus learned scale weights. Keep a validation-only selection rule and a fixed test. Report per-scan and per-subject R, MSE, MAE, skill over the training-mean baseline, lagged cross-correlation, and uncertainty intervals resampled over people rather than frames. With five test participants, intervals will be coarse.

1.3.3 Stage 3 — ambitious architecture and data extensions

If the shared seven-ROI model is stable, expand toward all 64 DiFuMo targets and test ROI-query decoding, temporal sequence prediction, and a modest functional-connectivity auxiliary loss. In parallel, compare a montage-aware EEG encoder such as REVE or a factorized time-frequency encoder such as CBraMod against the existing LaBraM-initialized model, first with frozen features or parameter-efficient tuning. Restrict any self-supervised pretraining to training participants if the claim is subject generalization.

For external validation, NATVIEW is a practical next cohort because it provides simultaneous resting and visual/naturalistic EEG-fMRI plus physiology and eye tracking; preserve it as an external test set rather than mixing its subjects into random training folds. The NeuroBOLT public release is rest-focused; auditory-task claims require access to the separate task data, which the release directs users to request from the authors (official dataset card, NATVIEW dataset paper). Leave-one-site-out testing, asynchronous EEG/fMRI prediction, clinical transfer, and broad montage robustness all require additional data and should not be claimed from the present split.

1.4 4. Cutting-edge literature organized by research idea

1.4.1 Shared regional decoders should test network structure as well as each time point

The closest new architectural ideas are multi-region prediction and ROI-conditioned decoding. **UnEBOLT is a peer-reviewed MIDL 2026 paper** that replaces separate region-specific models with a unified adaptive multi-region decoder and reports comparable ROI prediction while reconstructing subject-specific functional connectivity. **MARBLE is a peer-reviewed MICCAI 2026 paper** combining a Mamba–Transformer encoder with ROI-conditioned cross-attention; it reports lower parameter count and peak memory alongside BOLD and FC evaluation. A second MICCAI 2026 paper takes a different target: it predicts functional-connectivity graphs through semi-supervised latent alignment, not moment-by-moment BOLD. These are strong reasons to extend the current scalar-target path, but the connectivity paper's graph-level task must not be presented as direct evidence for frame-level regression (UnEBOLT, MARBLE, semi-supervised FC translation).

The direct code experiment is modest: enable the existing model's multi-output head for seven targets, share the EEG encoder, and evaluate per-ROI errors plus scan-wise signed FC similarity. If this works, add learned ROI embeddings or cross-attention queries; only then test an auxiliary FC loss computed over whole held-out scans or non-overlapping blocks. A shared head could regularize a small dataset, but it can also let high-variance or easy ROIs dominate. Use per-ROI losses or validation-chosen loss scaling, report negative transfer explicitly, and keep pointwise MSE as a control. The full 64-ROI version is a follow-up, not the first joint model.

1.4.2 Make neural-to-hemodynamic delay learnable, constrained, and inspectable

The current model maps a 16-second EEG window to one BOLD frame, then mean-pools internal representations. This leaves delay and regional response shape implicit. Earlier transfer-function analyses found that EEG spectral composition matters for BOLD prediction and developed data-driven mappings that avoid imposing one canonical HRF (Rosa et al., Sato et al.). Simultaneous EEG-fMRI studies also motivate testing region- and state-dependent coupling instead of assuming a universal lag (Butler et al.). Recent MARBLE uses a Mamba temporal component, but its authors frame this as an inductive bias for extended temporal dependencies; its Mamba component does not itself establish a physiologically correct HRF.

A feasible test is a validation-selected lag/window grid, followed by one interpretable temporal head: band-power/envelope features convolved with a small FIR basis, or a shallow temporal convolution over the existing encoder tokens. Compare those to current one-frame regression with identical splits. Plot ROI-wise lag curves and report where the peak correlation occurs, not only the maximum. The M3-scale experiments are tractable, but choosing lag and window length from the held-out test would overfit the five test people. Canonical HRF derivatives or causal temporal kernels are reasonable controlled alternatives; unconstrained per-person HRF estimation is a high-risk use of this small cohort.

1.4.3 Test geometry-aware and task-aligned EEG representations rather than model size

NeuroBOLT's two branches use learned channel-index embeddings; they do not directly encode electrode geometry. **REVE is peer-reviewed at NeurIPS 2025** and uses electrode xyz coordinates plus time in its positional representation, with broad pretraining and arbitrary-montage design. **CBraMod is peer-reviewed at ICLR 2025** and separates spatial and temporal attention. **TFM-Tokenizer is peer-reviewed at ICLR 2026** and learns discrete time-frequency motifs from single-channel EEG. Their results motivate transfer experiments, not a claim that any will improve fMRI regression. Much of their headline evidence is EEG classification or decoding, a different target from continuous BOLD regression (REVE, CBraMod, TFM-Tokenizer).

Start with a frozen REVE encoder plus the same regression head; then compare last-block tuning or LoRA with full fine-tuning. This isolates representation value and limits overfitting. For montage robustness, compare current channel IDs with coordinate embeddings or remove a prespecified subset of channels; actual cross-device transfer still needs another montage or cohort. Keep a compact ridge or EEGNet control. The peer-reviewed 2026 EEG-FM-Bench reports that model/data scale alone does not explain transfer and that multitask training can regularize data-scarce EEG tasks while sometimes causing negative transfer. A separate ICLR 2026 comparison found fine-tuned foundation models can fail to beat compact or classical decoders in data-scarce settings. These findings argue for matched protocols and not for importing a giant model on the basis of scale (EEG-FM-Bench, ICLR 2026 comparison).

1.4.4 Adapt the existing spectral branch before adding a second large encoder

The code already computes magnitude STFT at 0.5-, 1-, 2-, and 4-second windows and sums the four projected scales with equal weight. The most informative incremental test is therefore learned scale fusion, not another generic spectral pathway. Compare equal sum, global softmax scale weights, input-conditioned gates, each scale alone, and optionally overlapping windows at coarse scales. Also test log-power or band-envelope features against current magnitude features. CBraMod's factorized spatial/temporal design and CSBrain's cross-scale regional tokens provide design precedents; **CSBrain is peer-reviewed at NeurIPS 2025**, but its classification/task benchmarks do not prove a BOLD-regression gain. TFM's learned motifs are a separate front-end alternative, and can be frozen to reduce training cost.

The experiments touch models/model_multiscale.py where scale features are summed. Preserve the current four-scale implementation as the control, expose the scale count in configuration, and record parameter count, GPU time, and peak memory. A learned gate adds little compute; overlap and a new encoder add more. With 13 training participants, prefer a few predeclared variants over a broad hyperparameter search.

1.4.5 Diffusion and latent self-supervision are later-stage ideas

CATD, a **peer-reviewed IEEE Transactions on Medical Imaging 2025** method, uses condition-aligned latent blocks and temporal diffusion for EEG-to-fMRI generation. It is a relevant probabilistic direction, but reported temporal super-resolution conditions on observed low-resolution fMRI; that does not establish recovery of unobserved high-frequency BOLD from EEG alone. Begin instead with an ensemble or per-ROI mean/variance head and report calibration and interval coverage over held-out subjects/blocks. Explore diffusion only after deterministic baselines, external evaluation, and the uncertainty metric are in place (CATD).

EEG-JEPA and a 2026 masking-geometry comparison are **preprints**, not settled evidence. Their latent-target and mask-structure ideas could support a later self-supervised experiment, but the NeuroBOLT cohort is far smaller than large EEG pretraining corpora. If tested, pretrain only on training-subject EEG and compare waveform reconstruction, spectral reconstruction, and latent prediction with the same backbone and mask design; explicitly label use of held-out EEG during pretraining as transductive. The newer 2026 peer-reviewed EEG-FM-Bench similarly cautions that pretraining objective alignment and adaptation compatibility matter more than parameter count alone (EEG-JEPA preprint, mask-geometry preprint, EEG-FM-Bench).

1.5 5. First experiments on M3, then work that needs more data

Corrected paper-parity run
Hypothesis: Main-setting EEG filtering and β_2 may change results enough to matter for reproduction.
Minimum design: After the active jobs finish, rerun the seven cross-subject ROI models without extra EEG high-pass and with β=(0.9, 0.99); preserve the 18/5/6 subject-grouped split, seed, epoch schedule, and scan-aware evaluator. Follow with one-factor runs if attribution matters.
Primary readouts: Each held-out scan's R/MSE/MAE, macro mean and SD across scans and subjects, validation-selected epoch.

Simple baseline ladder
Hypothesis: NeuroBOLT should beat a transparent EEG predictor to justify its extra complexity.
Minimum design: Training-mean/intercept; ridge on lagged raw EEG; ridge on log band-power/envelopes; optional compact EEGNet. Tune regularization on validation only. Same targets, split and preprocessing.
Primary readouts: Scan-wise and subject-wise metrics; skill over mean; compute time.

Lag/window ablation
Hypothesis: The mapping depends on input history and target offset, and a small lag model may expose that dependence.
Minimum design: Prespecify lookbacks such as 8/16/24/32 seconds and a modest target offset grid; use validation to select, test once. Add an FIR or shallow temporal-convolution head only after the fixed-window grid.
Primary readouts: R/MSE/MAE, lagged cross-correlation curves, stability of ROI lag profile across held-out scans.

Shared seven-ROI model
Hypothesis: Shared features may reduce seven-model cost and preserve cross-region structure.
Minimum design: Enable multi-target loading; compare seven independent heads/models with one shared encoder and seven-output head. Keep architecture and optimizer otherwise fixed.
Primary readouts: Per-ROI scores, overall fit time, signed FC correlation and connectivity error computed per scan.

Spectral scale fusion
Hypothesis: Different EEG time scales may contribute unequally to slow BOLD predictions.
Minimum design: Compare current equal sum with learned global weights, conditional gate, individual scales, and optionally log-power. One change per run.
Primary readouts: Same scan-aware prediction metrics plus weights, parameter count and runtime.

Strict temporal-holdout check
Hypothesis: Full-scan filtering/normalization can make chronological scores optimistic for a prospective claim.
Minimum design: Separate intra-scan result; estimate target scaling from the train segment only and use causal or split-isolated filtering. Keep the paper-style full-scan path as a distinct protocol.
Primary readouts: Temporal test performance, sensitivity to purge gap, and clear protocol labels.

Use the 29 scans for the current data experiments, but remember that the independent unit is 22 participants and the final split contains only five test participants. Add subject-level bootstrap intervals cautiously, and treat seven-ROI or lag-family comparisons as exploratory unless primary metrics and correction rules are fixed in advance. Evaluate whether EEG alignment matters with scan-wise circular-shift or phase-randomized nulls. If physiology or motion traces are available, compare nuisance-only, EEG-only, and EEG-plus-nuisance baselines; predictive association alone does not establish neural causality.

The following questions are **out of scope for the currently staged M3 data**. Auditory task-to-rest or rest-to-task transfer needs the task-condition recordings and a defined event/frame alignment; the public release card says those data are available by request. External-site generalization needs a second cohort, for which NATVIEW is promising but differs in montage, scanner, tasks, and preprocessing. Asynchronous EEG-to-fMRI, patient or clinical transfer, and robust cross-montage performance require data beyond this one-site eyes-closed resting cohort. NATVIEW supplies 22 participants, simultaneous rest and visual/naturalistic runs, and physiological and eye-tracking data, making it a useful external test and confound-analysis set, not a large definitive replication cohort (NATVIEW paper, official NATVIEW page).

1.6 6. Limitations and primary references

The code findings above are repository-audit facts, not fresh scores or a claim that any proposed model will improve performance. The M3 jobs are left alone. The source cohort has a single site, eyes-closed rest, 22 participants, and only five held-out people in the supplied split; frame count is not independent sample size. The available notes do not establish that all staged scans include physiological or motion covariates. Recent 2025–2026 EEG encoders mostly report classification or decoding results, so they motivate adapters and ablations but do not supply regression evidence. Recent direct EEG-to-BOLD papers are highly relevant, but their splits, target definitions, preprocessing, and access to data/code must be checked before numerical comparisons.

Primary references used above include the NeuroBOLT paper, NeuroBOLT repository, UnEBOLT, MIDL 2026, MARBLE, MICCAI 2026, EEG-FM-Bench, ICML 2026, REVE, NeurIPS 2025, CBraMod, ICLR 2025, TFM-Tokenizer, ICLR 2026, CATD, IEEE TMI 2025, the NATVIEW dataset paper, and the NeuroBOLT dataset card. Recent preprint leads are explicitly labeled above rather than treated as established methods.

1.7 Conclusion

The project's next step is to make each result answer one clear question: paper parity, prospective temporal prediction, or cross-subject generalization. Once those protocols are separated and test evaluation is made scan-aware, a shared multi-region decoder and an inspectable lag model offer the strongest combination of scientific value and code fit. New encoders and generative methods should be judged against simple EEG baselines on the same held-out people; their broader EEG benchmark success alone cannot settle whether they predict BOLD better.